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BATTERY PACK

Abstract

Provided is a battery pack including a housing and a battery module inside the housing, wherein the housing includes a housing protrusion in the housing and housing ribs extending in different directions from each other from the housing protrusion, and wherein the housing ribs formed in any one housing protrusion may be apart from each other along a circumferential direction of the housing protrusion.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] The present application claims priority to and the benefit of Korean Patent Application No. 10-2024-0031524, filed on Mar. 5, 2024, in the Korean Intellectual Property Office, the entire disclosure of which is incorporated herein by reference.

BACKGROUND

1. Field

[0002] One or more embodiments relate to a battery pack.

2. Description of the Related Art

[0003] Secondary batteries are designed to be charged and discharged and may be used as an energy source in a mobile device, an electric vehicle, a hybrid vehicle, an electric bicycle, an uninterruptible power supply, etc. Depending on the types of external devices to which the secondary batteries are connected, the secondary batteries may be used in the form of a single battery cell or in the form of a module or pack in which a plurality of battery cells are collected as one unit.

[0004] The information described in the background art is only to improve the understanding of the background art of the present disclosure, and thus, may include information that is not included in the prior art.

SUMMARY

[0005] The present disclosure relates to various embodiment of a battery pack in which the durability and reliability of the battery pack may be improved by distributing a load of a battery module and enhancing a rigidity of a rib supporting the battery module.

[0006] However, the technical problem to be solved by the present disclosure is not limited to the above and other problems that are not mentioned above may be clearly understood by one of ordinary skill in the art from the detailed description below.

[0007] Additional aspects will be set forth in part in the description which follows and, in part, will be apparent from the description, or may be learned by practice of the presented embodiments of the present disclosure.

[0008] According to one or more embodiments, a battery pack includes a housing and a battery module accommodated in the housing. The housing includes a housing protrusion at the housing and housing ribs extending in different directions from each other from the housing protrusion. The housing ribs at the housing protrusion may be apart from each other along a circumferential direction of the housing protrusion.

[0009] The housing may include a first housing surface including a bottom surface of the housing and a second housing surface extending along an edge of the first housing surface in a height direction. Each of the housing ribs may be connected to the second housing surface.

[0010] Each of the housing ribs extending from the housing protrusion may be connected to another adjacent housing protrusion.

[0011] The housing protrusion may have a hollow cylindrical shape and may be connected to at least a portion of a bottom surface of the battery module.

[0012] The battery module may include a first coupling protrusion on the bottom surface of the battery module and the first coupling protrusion may be inserted into the housing protrusion.

[0013] The housing protrusion may include a first housing boss and the first coupling protrusion may be inserted into the first housing boss.

[0014] The housing may include a first fastening hole in a bottom surface of the housing and the battery module may include a second fastening hole in the bottom surface of the battery module.

[0015] The housing protrusion may include a second housing boss and the second fastening hole may be inserted into a third housing boss.

[0016] The housing protrusion may include a first housing boss, a second housing boss, and a third housing boss.

[0017] The first housing boss and the third housing boss may be inserted into the battery module and the second housing boss may be apart from a bottom surface of the battery module.

[0018] The second housing boss may be in a center portion of the housing in a width direction of the housing.

[0019] The third housing boss may include third housing bosses and the third housing bosses may correspond to a longitudinal side of the battery module.

[0020] The first housing boss corresponding to the battery module may be inside a first area partitioned by third housing bosses corresponding to the battery module and the housing ribs may connect the third housing bosses together.

[0021] The first housing boss may include first housing bosses and the first housing bosses may correspond to a center portion of the battery module.

[0022] The battery module may include two or more battery modules and the second housing boss may include second housing bosses.

[0023] The second housing bosses may be in a second area partitioned by the third housing bosses corresponding to an outer longitudinal side of one battery module, the third housing bosses may correspond to an outer longitudinal side of another battery module, and the housing ribs may connect the third housing bosses together.

[0024] The first area may include first areas and the second area may be between the first areas and may be in contact with each of the first areas.

[0025] The housing rib may include a first housing rib extending in a horizontal direction, a second housing rib extending in a vertical direction, and a third housing rib extending in a diagonal direction between the first housing rib and the second housing rib.

[0026] The first housing rib, the second housing rib, and the third housing rib may be connected to the second housing surface.

[0027] A height of the housing ribs may be less than a height of the housing protrusion.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0028] The above and other aspects, features, and advantages of certain embodiments of the present disclosure will be more apparent from the following description taken in conjunction with the accompanying drawings, in which:

[0029] The accompanying drawings are for illustrating example embodiments and provide understanding of the technical concept of the present disclosure together with the detailed description described below. The present disclosure is not construed as limited to the details of the description of the drawings.

[0030] FIG. 1 is a perspective view of a battery pack;

[0031] FIG. 2 is an exploded view of a battery pack;

[0032] FIG. 3 shows an inside of a housing;

[0033] FIG. 4 shows an enlarged portion of FIG. 3;

[0034] FIG. 5 is a plan view of FIG. 4;

[0035] FIG. 6 is a cross section of FIG. 3 taken along line VI-VI';

[0036] FIG. 7 is a cross section of FIG. 3 taken along line VII-VII';

[0037] FIG. 8 shows another enlarged portion of FIG. 3;

[0038] FIG. 9 shows a state in which a battery module is coupled to a housing;

[0039] FIG. 10 shows another enlarged portion of FIG. 3;

[0040] FIG. 11 is a plan view of FIG. 10;

[0041] FIG. 12 shows an inner cover and a battery module;

[0042] FIG. 13 is an inner surface of an inner cover; and

[0043] FIG. 14 shows an enlarged portion of FIG. 13.

DETAILED DESCRIPTION

[0044] Reference will now be made in detail to embodiments, examples of which are illustrated in the accompanying drawings, wherein like reference numerals refer to like elements throughout. In this regard, the present embodiments may have different forms and should not be construed as being limited to the descriptions set forth herein. Accordingly, the embodiments are merely described below, by referring to the figures, to explain aspects of the present description. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items. Expressions such as “at least one of,” if preceding a list of elements, modify the entire list of elements and do not modify the individual elements of the list.

[0045] Some embodiments and methods therefor in the present disclosure may be understood by referring to the detailed descriptions and drawings of the embodiment. The embodiments described have a variety of modifications, may be implemented in different forms, and are not limited to the embodiments described herein. The features of the various embodiments may be combined in part or in whole with each other. The embodiments each may be implemented independently or in relation to each other. The described embodiments are provided as examples to ensure that the present disclosure is and becomes complete and to completely convey the idea of the present disclosure to those of ordinary skill in the art to which the present disclosure belongs. All changes, equivalents, and substitutes that do not depart from the spirit and technical scope of the present disclosure are encompassed in embodiments. Thus, processes, elements, and technologies that are not necessary for a complete understanding of the embodiment by one of ordinary skill in the art may not be explained.

[0046] Unless separately mentioned otherwise throughout the accompanying drawings and specification, the same reference number, character, or a combination thereof indicates the same component, and thus, overlapping descriptions are omitted. Parts that are irrelevant to the description are omitted to clearly describe the present disclosure.

[0047] The relative size of the elements, layers, and areas in the drawing may be exaggerated for clarity. The use of hatching and/or shades in the accompanying drawings may be provided generally to clarify the boundaries between adjacent elements. Therefore, the presence of hatching or shades do not represent certain materials, material characteristics, dimensions, rates, commonalities between drawing elements, and/or other characteristics of an unspecified element, properties, appropriate forms or requirements for properties, etc.

[0048] Various embodiments are described herein with reference to the cross-sectional examples, which are schematic examples of embodiments and/or intermediate structures. Thus, for example, the shape of the drawing may be changed as a result of manufacturing technology and/or tolerance. The specific structural or functional explanations in the present disclosure are only examples for explaining embodiments according to the concept of the present disclosure. Therefore, the embodiments disclosed herein should not be construed as being limited to a shape of an illustrated area and may include, for example, deviations in shape due to manufacturing.

[0049] The area shown in the drawing may be essentially schematic and the shape of the area is not intended to illustrate the actual shape of the device area nor limit the area. As recognized by one of ordinary skill in the art, the described embodiments may be modified in various ways within the limit of the idea or scope of the present disclosure.

[0050] Many specific details are provided for a complete understanding of various embodiments in the present disclosure. However, various embodiments may be implemented without the specific details or with one or more of the specific details. In some embodiments, well-known structures and devices may be illustrated in the form of block diagrams to avoid unnecessary ambiguity in various embodiments.

[0051] As shown in the drawing, to make it easier to explain the relationship between one element or feature and other elements or features, spatially relative terms such as “below,” “above,”

“lower,” “upper”, etc. may be used. Spatially relative terms are intended to include various directions of devices in use or in operation, in addition to the directions illustrated in the drawing. For example, if the device in the drawing is turned upside down, elements or features described as being “below” or “lower” may be “above” other elements or features. Therefore, as an exemplary term, “below” and “lower” may include both upward and downward directions. The device may face another direction (e.g., a direction rotated by 90 degrees or another direction) and the spatially relative explanation used herein should be construed accordingly. Similarly, if a first portion is disposed “above” a second portion, this means that the first portion is disposed at the upper side or lower side of the second portion.

[0052] The expression “viewed in a plane” refers to viewing an object from above and the expression “in a schematic cross-sectional view” refers to vertically cutting the object portion to obtain a schematic cross-section. The term “viewed from the side” refers to a first object being above, below, or on the side of a second object and vice versa. The term “overlapping” may include the meaning of a layer, layering, surface, extension, covering, partial covering, or any other appropriate term that is or may be understood by one of ordinary skill in the art. The expression “not overlapping” may include the meaning of “apart from ~” and other appropriate equivalents that may be recognized and understood by one of ordinary skill in the art. The term “surface” may mean that a first object may directly or indirectly face a second object. If there is a third object between the first object and the second object, the first object and the second object may be understood as facing each other, but as indirectly facing each other.

[0053] If an element, layer, region, or component is referred to as being “formed in,” “connected to,” or “coupled to” another element, the element, layer, region, or component, the element, layer, region, or component may be directly formed in, indirectly formed in, connected to, or coupled to another element, layer, region, or component. The above terms may collectively refer to direct or indirect coupling or connection and integral and non-integral coupling or connection of elements, layers, areas, or components such that one or more elements, layers, regions, or components are present. For example, if elements, layers, areas, or components are referred to as “electrically connected” or “electrically coupled” to other elements, layers, areas, or components, the elements, layers, areas, or components may be directly electrically connected or coupled to the other elements, layers, areas, or components or intervening elements, layers, areas, or components may be present. However, “direct connection” or “direct coupling” may refer to one component being directly connected or coupled to other components without intermediate components. If a portion of a layer, film, area, guide plate, etc. is formed on another portion, the formation direction is not limited to an upper direction and the portion may be formed on the side or a lower portion. In some embodiments, if a portion of the layer, film, area, guide plate, etc. are formed in the “bottom” of another portion, the portion may not only be “right below” the other portion but another portion may be between the portion and the other portion. In some embodiments, other expressions that describe the relationship between the components, such as “between,” “right in between,” “adjacent to ~,” or “directly adjacent to” may be construed similarly. If the element or layer is mentioned as being “between” two elements or layers, the element may be the only element between the two elements or layers or another element may be between the two elements or layers.

[0054] For the purpose of the present disclosure, expressions such as “at least one” or “any one” do not limit the order of individual elements. For example, “at least one of X, Y, and Z,” “at least one of X, Y, or Z,” and “at least one selected from a group including X, Y, and Z” may include X alone, Y alone, Z alone, and any combination of at least two of X, Y, and Z. Similarly, expressions such as “at least one of A and B” and “at least one of A or B” may include A, B, or A and B. The term “and/or” herein generally includes all combinations of one or more in a list of related items. For example, expressions such as “A and/or B” may include A, B, or A and B.

[0055] Terms such as “first,” “second,” “third,” etc. may be used to explain various elements, components, areas, layers, and/or cross sections, but these elements, components, areas, layers,

and/or cross sections are not limited by these terms. The terms are used to distinguish one element, component, area, layer, or cross section from other elements, components, areas, layers, or cross sections. Thus, a first element, a first component, a first area, a first layer, or a first cross section described below may be referred to as a second element, a second component, a second area, a second layer, or a second cross section within the concept and scope of the present disclosure. Describing an element as a “first” element may not require or suggest the presence of a second element or other elements. Terms such as “first,” “second,” etc. may be used herein to distinguish different categories or element sets. For clear description, terms such as “first,” “second,” etc. may represent “a first category (or first set),” and “a second category (or second set),” respectively.

[0056] The terms used herein are used to describe certain embodiments and are not intended to limit the present disclosure. As used herein, singular forms are intended to include plural forms and plural forms are intended to include singular forms unless clearly described otherwise in context. The terms “including,” “provided with,” and “having” refer to specifying the presence of features, integers, and steps that are used herein. The terms do not exclude the presence or addition of one or more other functions, steps, operations, components and/or a group thereof.

[0057] If one or more embodiments may be implemented differently, a specific process order may be performed differently from the described order. For example, two consecutively described processes may be performed substantially at the same time or performed in an order opposite to the described order.

[0058] The terms “substantially,” “about,” “approximately,” and similar terms are used as approximations rather than as terms of degree and refer to satisfying an inherent deviation (e.g., a deviation scope caused by limitations of a measurement system) scope of measured or calculated values. For example, “about” may refer to a range within one or more standard deviations or within $\pm 30\%$, 20% , 10% , and 5% of specified values.

[0059] Unless defined otherwise, all terms used herein, including technical or scientific terms, have the same meaning as commonly understood by one of ordinary skill in the technical field to which the present disclosure belongs. General terms such as terms that are previously defined in the dictionary are to be construed as having meanings that match the related technology and/or context of the specification and, unless defined explicitly, are not construed as having ideal or overly formal meanings.

[0060] FIG. 1 is a perspective view of a battery pack 10, FIG. 2 is an exploded view of the battery pack 10, FIG. 3 shows the inside of a housing 100, and FIG. 4 shows an enlarged portion of FIG. 3. FIG. 4 shows an enlarged upper left corner portion of FIG. 3. FIG. 5 is a plan view of FIG. 4, FIG. 6 is a cross section of the housing 100 of FIG. 3 taken along cross-sectional line VI-VI', and FIG. 7 is a cross-section of the housing 100 of FIG. 3 taken along cross-sectional line VII-VII'. FIG. 6 shows a cross section of the housing 100 and a battery module 400 that are coupled to each other.

[0061] FIG. 8 shows another enlarged portion of FIG. 3. FIG. 8 shows an enlarged second housing surface 102 of the housing 100 on which a contact protrusion 150 is formed. FIG. 9 shows a state in which the battery module 400 is coupled to the housing 100. FIG. 9 is an enlarged plan view of the surrounding of the contact protrusion 150. FIG. 10 shows another enlarged portion of FIG. 3. FIG. 10 shows an enlarged second housing surface 102 of the housing 100 on which a pocket 160 is formed. FIG. 11 is a plan view of FIG. 10, FIG. 12 shows an inner cover 300 and a battery module 400, FIG. 13 shows an inner surface of the inner cover 300, and FIG. 14 is an enlarge portion of FIG. 13. FIG. 14 shows an enlarged upper left corner portion of FIG. 13.

[0062] The battery pack 10 may be applied to large applications such as energy storage systems (ESS) or electric vehicles or small and medium-sized applications such as electric tools or smartphones. The battery pack 10 may be coupled to a different device. In one or more embodiments, the battery pack 10 may be connected to various external devices through an adaptor A. The external device may be electric tools such as a chain saw, a drill, a lawn mower, etc.

[0063] The battery pack 10 may be embodied in a form by which users may wear or carry. In one

or more embodiments, the battery pack **10** may be a backpack configured to be worn by the user. A connection member such as a string, buckle, or belt may be attached to the battery pack **10** (e.g., the back of the battery pack **10**). The user may connect the connection member to the body such as the shoulder, waist, and/or back to carry the battery pack **10** and to connect the battery pack **10** to an external device such as an electric tool, to supply power to external devices. The battery pack **10** may include a handle **H** for the user to carry around. In FIG. **1**, the handle **H** is illustrated as being on the upper portion of the housing **100**, but one or more handles **H** may be provided in the front surface of the housing **100** (e.g., a surface facing a Z-axis direction of FIG. **1**), the rear surface, side surface, bottom surface, or other surfaces of the housing **100**.

[0064] The battery pack **10** may include the housing **100** and at least one battery module **400** accommodated in the housing **100**. The housing **100** may include one or more housing protrusions **110** in the housing **100** and housing ribs **120** extending from the housing protrusion **110** in different directions from each other. The housing rib **120** in any one housing protrusion **110** may be apart from each other along a circumferential direction of the housing protrusion **110**.

[0065] The battery pack **10** may include a housing **100**, an external cover **200**, an inner cover **300**, and a battery module **400**.

[0066] The housing **100** may retain and/or support other components (e.g., the external cover **200**, the inner cover **300**, and the battery module **400**) of the battery pack **10**. The housing **100** may be in the lower portion of the battery pack **10** (e.g., the lower portion in the Z-axis direction of FIG. **1**) and the interior thereof may be empty (e.g., the housing **100** may define an open compartment). The battery module **400** may be inside the housing **100**. The housing **100** may have a rectangular shape in which the inside is empty and the upper surface is open. However, the shape of the housing **100** is not limited thereto and may vary according to the shape of the battery pack **10** and/or the size and number of the battery module(s) **400**.

[0067] The housing **100** may include a first housing surface **101** and a second housing surface **102**.

[0068] The first housing surface **101** may include a bottom surface of the housing **100**. In one or more embodiments, as shown in FIG. **3**, the first housing surface **101** may correspond to the inner bottom surface of the housing **100** and at least a portion of the protrusion **110**, the housing rib **120**, a mounting guide **130**, a contact protrusion **150**, and a pocket **160** may be on the first housing surface **101**. The battery module **400** may be on the first housing surface **101**.

[0069] The second housing surface **102** may include a side surface of the housing **100**. In one or more embodiments, as shown in FIG. **3**, the second housing surface **102** may extend along an edge (e.g., a periphery) of the first housing surface **101** and correspond to an inner side of the housing **100**. The second housing surface **102** may extend from an edge of the first housing surface **101** in a height direction (e.g., the Z-axis direction of FIG. **1**) and, together with the first housing surface **101**, may partition or bound an inner space in which the battery module **400** is arranged. At least a portion of each of the housing rib **120**, the mounting guide **130**, the contact protrusion **150**, and the pocket **160** may be on the second housing surface **102**.

[0070] The housing **100** may include the housing protrusion **110** and the housing rib **120**.

[0071] At least one housing protrusion **110** may be in the housing **100** and may support the battery module **400**. together with the first housing surface **101**, as shown in FIG. **3**, a plurality of housing protrusions **110** may be on the first housing surface **101** and may protrude upwards (e.g., Z-axis direction) from the first housing surface **101**. The housing protrusion **110** may support the battery module **400** by being in contact with the bottom surface of the battery module **400**. The bottom surface of the battery module **400** may be spaced apart by a gap from the first housing surface **101** by the housing protrusion **110**, thereby forming a space in which heat generated from the lower portion of the battery module **400** is discharged. The housing protrusion **110** may have a hollow cylinder shape. A portion of the battery module **400** (e.g., a first coupling protrusion **420** as shown in FIG. **6**) may be inserted into the housing protrusion **110** and be fixed to the housing **100**. The plurality of housing protrusions **110** may be spaced apart from each other and connected to each

other through the housing rib **120**. The housing protrusion **110** may be connected to at least a portion of the bottom surface of the battery module **400**.

[0072] The housing protrusion **110** may include a first housing boss **111**, a second housing boss **112**, and a third housing boss **113**.

[0073] At least one housing boss **111** may be on the first housing surface **100** and be connected to the battery module **400**. In one or more embodiments, the first housing boss **111** may include a plurality of first housing bosses **111**. A plurality of first housing bosses **111** may correspond to any one battery module **400** and at least a portion of the battery module **400** may be inserted into the first housing boss **111** or at least a portion of the first housing boss **111** may be inserted into the battery module **400** such that the battery module **400** is connected to the first housing boss **111**. In one or more embodiments, the first coupling protrusion **420** of the battery module **400** may be inserted into the first housing boss **111**. In one or more embodiments, as shown in FIG. 6, the battery module **400** may include the first coupling protrusion **420**. The first coupling protrusion **420** may extend downwards (e.g., in the Z-axis direction) from the bottom surface of the battery module **400** and have a cylinder shape. The first coupling protrusion **420** may have a height $h2$ lower than a height $H1$ of the first housing boss **111**. The height $H1$ may be a height from the first housing surface **101** to the upper end of the first housing boss **111** and the height $h2$ may be a height from the bottom surface of the battery module **400** to a lower end of the first coupling protrusion **420**. The height $H1$ may be greater than the height h (see FIG. 4) of the housing rib **120**. The first coupling protrusion **420** may have an outer diameter that is the same as or less than an inner diameter of the first housing boss **111** and the first coupling protrusion **420** may be inserted into the first housing boss **111**.

[0074] The first housing boss **111** may correspond to a center portion of the battery module **400**. In one or more embodiments, as shown in FIG. 3, two first housing bosses **111** may be formed in a width direction (e.g., the X-axis direction of FIG. 3) of the battery module **400** and in the center portion of the battery module **400** (shown in dashed lines). The two first housing bosses **111** may be formed on the same line in a length direction (e.g., the Y-axis direction of FIG. 3) of the housing **100**. A plurality of first housing bosses **111** may correspond to each battery module **400**. In one or more embodiments, as shown in FIG. 3, two first housing bosses **111** may correspond to each battery module **400** and a total of four first housing bosses **111** may be on the first housing surface **101**. However, the number of the first housing boss **111** is not limited to four and may vary according, for example, to the size and shape of the battery module **400** and the battery pack **10**.

[0075] The first housing boss **111** may be connected to the plurality of housing ribs **120**. In one or more embodiments, as shown in FIGS. 3 and 4, the plurality of housing ribs **120** may be connected to the outer circumferential surface of the first housing boss **111**. The plurality of housing ribs **120** (e.g., a first housing rib **121**, a second housing rib **122**, and a third housing rib **123**) may be connected along the circumference of the first housing boss **111** at equal or different angular intervals. The plurality of housing ribs **120** may extend radially outward from the center of the first housing boss **111**. The radially extending plurality of housing ribs **120** may be connected to another first housing boss **111**, a second housing boss **112**, and/or a third housing boss **113**, or the second housing surface **102**. Therefore, the first housing boss **111**, the second housing boss **112**, and the third housing boss **113** may be connected to each other by a plurality of housing ribs **120**, thereby efficiently distributing the weight of the battery module **400** and enhancing the rigidity of the first housing boss **111**, the second housing boss **112**, and the third housing boss **113**.

[0076] The first housing boss **111** may be located more toward the inside or interior than the third housing boss **113**. In one or more embodiments, the first housing boss **111** may be in an area formed by connecting the plurality of third housing bosses **113**. In one or more embodiments, as shown in FIG. 3, two first housing bosses **111** may be inside area A1 and area A2 (e.g., the first area) formed by connecting the plurality of third housing bosses **113** corresponding to each battery module **400**. As shown in FIGS. 3 and 5, the area A1 and the area A2 may be virtual areas formed

by drawing imaginary lines connecting the plurality of third housing bosses **113** (e.g., six third housing bosses **113**) corresponding to any one battery module **400** to the first housing rib **121** and the second housing rib **122**, which are connected to each of the third housing bosses **113**. The first housing boss **111** corresponding to any one battery module **400** may be inside the first area (area **A1** and area **A2**) that is partitioned with the plurality of third housing bosses **113** and the housing ribs **120** connecting the plurality of third housing bosses **113**.

[0077] Therefore, the first housing boss **111** may first be connected to the battery module **400** such that the housing **100** is firstly connected to the battery module **400** more toward the inside or the interior than the plurality of third housing bosses **113**. In this configuration, a fastening member such as the bolt may fasten the third housing boss **113** to the battery module **400** in the outside of the first housing boss **111**, and thus, positions of the housing **100** and the battery module **400** may be easily aligned and the battery module **400** may be stably fastened to the housing **100**.

[0078] At least one second housing boss **112** may be on the first housing surface **101** and may not be directly connected to the battery module **400**. In one or more embodiments, the second housing boss **112** may include a plurality of second housing bosses **112**. The first housing boss **111** and the third housing boss **113** may be connected to the battery module **400** by inserting the battery module **400** or by a fastening member, and the second housing boss **112** may not be in contact with the battery module **400**. In a configuration in which the battery module **400** is connected to the first housing boss **111** and the third housing boss **113**, the second housing boss **112** may be spaced apart by a gap from the bottom surface of the battery module **400**. The second housing boss **112** may have the height **H2**. The height **H2** may be less than the height **H1** of the first housing boss **111** and less than the height **H3** of the third housing boss **113**. The height **H2** may be greater than the height **h** of the housing rib **120**.

[0079] The second housing boss **112** may be in the center portion (or substantially the center portion) of the housing **100**. In one or more embodiments, a plurality of second housing bosses **112** may be arranged in the width direction of the housing **100** (e.g., the X-axis direction of FIG. 3) in the center portion (or substantially the center portion) of the housing **100**. In one or more embodiments, two second housing bosses **112** may be formed in the center of the housing **100**. The two second housing bosses **112** may be arranged on the same line in the length direction (e.g., the Y-axis direction of FIG. 3) of the housing **100**. The two second housing bosses **112** may be under any one battery module **400** or the second housing bosses **112** may not overlap the battery module **400**. The plurality of second housing bosses **112** may connect the first housing boss **111** to the second housing boss **113**, the first housing boss **111**, and the second housing boss **113** corresponding to different battery modules **400**.

[0080] The second housing boss **112** may be located more toward the inside or the interior than the first housing boss **111** and the third housing boss **113**. In one or more embodiments, as shown in FIG. 3, two second housing bosses **112** may be inside area **A3** (a second area) formed by connecting the plurality of third housing bosses **113**. As shown in FIGS. 3 and 5, the area **A3** may be a virtual area formed by drawing imaginary lines connecting the plurality of third housing bosses **113** corresponding to different battery modules **400** from each other to the first housing rib **121** and the second housing rib **122**, which are connected to each of the third housing bosses **113**. The plurality of second housing bosses **112** may be in the second area (area **A3**) that is partitioned by the plurality of third housing bosses **113** corresponding to an outer longitudinal side of the battery module **400**, the plurality of third housing bosses **113** corresponding to the outer longitudinal side of the other battery module **400**, and the housing ribs **120** connecting the plurality of third housing bosses **113**. In one or more embodiments, the first region (area **A1** and area **A2**) may include a plurality of first areas and the second region (area **A3**) may be formed between the plurality of first areas and may be respectively in contact with the plurality of first areas. In one or more embodiments, as shown in FIG. 3, a longitudinal side of the area **A3** may be in contact with the inner longitudinal side of the area **A1** and the other longitudinal side of the area **A3** may be in

contact with the inner longitudinal side of the area A2.

[0081] The second housing boss **112** may be connected to the plurality of housing ribs **120**. In one or more embodiments, as shown in FIGS. **3** and **4**, the plurality of housing ribs **120** may be connected to the outer circumferential surface of the second housing boss **111**. The plurality of housing ribs **120** (e.g., the first housing rib **121**, the second housing rib **122**, and the third housing rib **123**) may be connected to the circumference of the second housing boss **112** at equal or different angular intervals. The plurality of housing ribs **120** may extend radially outward from the center of the second housing boss **112**. The radially extending plurality of housing ribs **120** may be connected to another second housing boss **112**, the first housing boss **111** and/or the third housing boss **113**, or the second housing surface **102**. The second housing boss **112** may be connected to the first housing boss **111** and the second housing boss **112** that correspond to different battery modules **400**, thereby distributing the load and vibration applied to the first housing boss **111** and the second housing boss **112**. In one or more embodiments, the second housing boss **112** may not be directly connected to the battery module **400** and may be connected to the first housing boss **111** and the third housing boss **113** through the housing rib **120**. In one or more embodiments, as shown in FIG. **3**, the second housing boss **112** may be connected to the first housing boss **111** adjacent thereto by the first housing rib **121** or the second housing boss **112** may be connected to the third housing boss **113** by the third housing rib **123**.

[0082] The second housing boss **112** may be on the same line as the first housing boss **111**. In one or more embodiments, the center portion of the second housing boss **112** may be on an imaginary line extending in the horizontal direction (e.g., the X-axis direction of FIG. **3**) from the center portion of the first housing boss **111**.

[0083] At least one third housing boss **113** may be on the first housing surface **101** and may be connected to the battery module **400**. In one or more embodiments, the third housing boss **113** may include a plurality of third housing bosses **113**. A plurality of third housing bosses **113** may correspond to any one battery module **400** and at least a portion of the battery module **400** may be inserted into the third housing boss **113** or at least a portion of the third housing boss **113** may be inserted into the battery module **400** such that the battery module **400** is connected to the third housing boss **113**. In one or more embodiments, a second fastening hole **430** of the battery module **400** may be inserted into the third housing boss **113**. In one or more embodiments, as shown in FIG. **7**, the battery module **400** may include the second fastening hole **430** on the bottom surface of the battery module **400** and the second fastening hole **430** may correspond to the third housing boss **113** and the first fastening hole **140** of the housing **100**. The first fastening hole **140** may be at the bottom surface of the housing **100** and may be in communication with the third housing boss **113**. In a configuration in which the first fastening hole **140** corresponds to the second fastening hole **430**, the lower end of the second fastening hole **430** may be inserted into the third housing boss **113**. The fastening member, such as the bolt or other fastener, may be inserted from below the first fastening hole **140** such that the housing **100** is coupled to the battery module **400**. The third housing boss **113** may have a height H3 from the first housing surface **101** to an upper end of the third housing boss **113**. The inner bottom surface of the third housing boss **113** may be above the first housing surface **101** and the lower end of the second fastening hole **430** may be in contact with the inner bottom surface of the third housing boss **113**. The third housing boss **430** may have an outer diameter that is the same as or less than an inner diameter of the third housing boss **113** and the third housing boss **430** may be inserted into the third housing boss **113**.

[0084] The third housing boss **113** may correspond to an edge portion of the battery module **400**. In one or more embodiments, as shown in FIG. **3**, three third housing bosses **113** may be formed to correspond to each edge (e.g., a pair of longitudinal sides) of the battery module **400** (shown in dashed lines). The plurality (e.g., three) of third housing bosses **113** corresponding to any one longitudinal side of the battery module **400** may be on the same imaginary line in the length direction (e.g., the Y-axis direction of FIG. **3**) of the housing **100**. In one or more embodiments, six

third housing bosses **113** may correspond to one battery module **400**, and a total of twelve third housing bosses **113** may be on the first housing surface **101**. However, the number of the third housing boss **113** is not limited to twelve and may vary, for example, according to the size and shape of the battery module **400** and the battery pack **10**.

[0085] The third housing boss **113** may be connected to the plurality of housing ribs **120**. In one or more embodiments, as shown in FIGS. **3** and **4**, the plurality of housing ribs **120** may be connected to the outer circumferential surface of the third housing boss **113**. The plurality of housing ribs **120** (e.g., the first housing rib **121**, the second housing rib **122**, and the third housing rib **123**) may be connected along the circumference of the third housing boss **113** at equal or different angular intervals. The plurality of housing ribs **120** may extend radially outward from the center of the third housing boss **113**. The radially extending plurality of housing ribs **120** may be connected to another third housing boss **113**, the first housing boss **111** and/or the second housing boss **112**, or the second housing surface **102**. Therefore, the third housing boss **111**, the first housing boss **111**, and the second housing boss **112** may be connected to each other by a plurality of housing ribs **120**, thereby efficiently distributing the weight of the battery module **400** and enhancing the rigidity of the first housing boss **111**, the second housing boss **112**, and the third housing boss **113**.

[0086] The third housing boss **113** may be located more to the outside than the first housing boss **111** and the second housing boss **112**. In one or more embodiments, as shown in FIG. **3**, the first housing boss **111** may be formed in each of the area **A1** and the area **A2** formed by connecting the plurality of third housing bosses **113** corresponding to each battery module **400** and the plurality of third housing bosses **113** may be formed on the outer edge of the area **A1** and the area **A2**.

[0087] The plurality of housing ribs **120** connected to the housing protrusion **110** may extend from different portions of the housing protrusion **110**. In one or more embodiments, as shown in FIG. **3**, the plurality of housing ribs **120** extending from the outer circumferential surface of any one housing protrusion **110** may extend from different portions of the corresponding housing protrusion **110** and may not overlap or be connected (or at least not directly connected) to adjacent housing ribs **120**. Therefore, the impact or vibration transmitted from any one housing rib **120** may be absorbed by the housing protrusion **110** and transmitted to another housing rib **120**, thereby increasing the durability of the housing rib **120**.

[0088] The housing rib **120** may be on the first housing surface **101** and/or the second housing surface **102** and may connect the plurality of housing protrusions **110** together. In one or more embodiments, the plurality of housing ribs **120** may extend along the first housing surface **101** from the circumference of the plurality of housing protrusions **110** and be connected to an adjacent housing protrusion **110**. The plurality of housing ribs **120** may cross each other to form a grid pattern or a net pattern. The housing ribs **120** may distribute the load and vibration of the battery module **400** applied to the housing protrusion **110**, thereby improving the durability of the housing **100** including the housing protrusion **110**.

[0089] The housing ribs **120** may be connected to the second housing surface **102**. In one or more embodiments, as shown in FIGS. **3** and **4**, the housing ribs **120** extending from the housing protrusion **110** adjacent or proximate to the second housing surface **102** of the housing **100** may be connected to the second housing surface **102**. Therefore, the load and vibration of the battery module **400** may be distributed through the second housing surface **102** through the housing protrusion **110** and the housing rib **120**.

[0090] The housing rib **120** may include the first housing rib **121**, the second housing rib **122**, and the third housing rib **123**.

[0091] A plurality of first housing ribs **121** may be on the first housing surface **101** and the second housing surface **102** and may be connected to the housing protrusion **110**. In one or more embodiments, the first housing rib **121** may extend in the width direction (e.g., the X-axis direction of FIG. **3**) of the housing **100** along the first housing surface **101**. The housing rib **121** may connect the first housing boss **111**, the second housing boss **112**, and the third housing boss **113** to each

other in a vertical direction (e.g., the Y-axis direction). The first housing rib **121** may have a height h and the height h may be less than the height $H1$ of the first housing boss **111**, less than the height $H2$ of the second housing boss **112**, and less than the height $H3$ of the third housing boss **113**. The first housing rib **121** may extend from the outer circumferential surface of the first housing boss **111**, the second housing boss **112**, and the third housing boss **113** and may cross the second housing rib **122**. The first housing rib **121** may be connected to the second housing surface **102**. In one or more embodiments, as shown in FIGS. **3** and **4**, a portion of the first housing rib **121** extending from the first housing boss **111**, the second housing boss **112**, and the third housing boss **113** may extend along the first housing surface **101** and may contact the second housing surface **102**. Therefore, the load and vibration transmitted along the first housing rib **121** may be distributed to the second housing surface **102**.

[0092] A plurality of second housing ribs **122** may be on the first housing surface **101** and the second housing surface **102** and may be connected to the housing protrusion **110**. In one or more embodiments, the second housing rib **122** may extend in the width direction (e.g., the Y-axis direction of FIG. **3**) of the housing **100** along the first housing surface **101**. The housing rib **122** may connect the first housing boss **111**, the second housing boss **112**, and the third housing boss **113** to each other in the second direction (e.g., the Y-axis direction of FIG. **3** or a vertical direction). The second housing rib **122** may have the same height h as the first housing rib **121**, and the height h may be less than the height $H1$ of the first housing boss **111**, less than the height $H2$ of the second housing boss **112**, and less than the height $H3$ of the third housing boss **113**. The second housing rib **122** may extend from the outer circumferential surface of the first housing boss **111**, the second housing boss **112**, and the third housing boss **113** and may cross the first housing rib **121** and the third housing rib **123**. The second housing rib **122** may be connected to the second housing surface **102**. In one or more embodiments, as shown in FIGS. **3** and **4**, a portion of the second housing rib **122** extending from the first housing boss **111**, the second housing boss **112**, and the third housing boss **113** may extend along the first housing surface **101** and contact the second housing surface **102**. Therefore, the load and vibration transmitted along the second housing rib **122** may be distributed to the second housing surface **102**.

[0093] A plurality of third housing ribs **123** may be on the first housing surface **101** and the second housing surface **102** and may be connected to the housing protrusion **110**. In one or more embodiments, the third housing rib **123** may extend in a diagonal direction (e.g., a direction between the X-axis and the Y-axis of FIG. **3**) of the housing **100** along the first housing surface **101**. The third housing rib **123** may connect the first housing boss **111**, the second housing boss **112**, and the third housing boss **113** to each other in a diagonal direction. The third housing rib **123** may have the same height h as the first housing rib **121**, and the height h may be less than the height $H1$ of the first housing boss **111**, less than the height $H2$ of the second housing boss **112**, and less than the height $H3$ of the third housing boss **113**. The third housing rib **123** may extend from the outer circumferential surface of the first housing boss **111**, the second housing boss **112**, and the third housing boss **113** and may cross the first housing rib **121** and the second housing rib **122**. The third housing rib **123** may be connected to the second housing surface **102**. In one or more embodiments, as shown in FIGS. **3** and **4**, a portion of the third housing rib **123** extending from the first housing boss **111**, the second housing boss **112**, and the third housing boss **113** may extend along the first housing surface **101** and contact the second housing surface **102**. Therefore, the load and vibration transmitted along the third housing rib **123** may be distributed to the second housing surface **102**.

[0094] The first housing rib **121**, the second housing rib **122**, and the third housing rib **123** extending from any one housing protrusion **110** may be connected to different portions of the outer circumferential surface of the corresponding housing protrusion **110**. In one or more embodiments, as shown in FIG. **3**, the first housing rib **121**, the second housing rib **123**, and the third housing rib **123** may be connected to the outer circumferential surface of the first housing boss **111**, the second

housing boss **112**, and the third housing boss **113**, respectively. The first housing rib **121**, the second housing rib **122**, and the third housing rib **123** extending from any one first housing boss **111**, second housing boss **112**, and third housing boss **113** may be connected to different portions of the circumferential surface of the corresponding first housing boss **111**, the second housing boss **112**, and the third housing boss **113**. Therefore, the impact or vibration transmitted from the first housing rib **121**, the second housing rib **122**, and the third housing rib **123** may be absorbed by the housing protrusion **110** and transmitted to another first housing rib **121**, another second housing rib **122**, and another third housing rib **123**, thereby enhancing the durability of the first housing rib **121**, the second housing rib **122**, and the third housing rib **123**.

[0095] The housing **100** may further include the mounting guide **130**. At least one mounting guide **130** may be on the first housing surface **101** and the second housing surface **102** and the mounting guide **130** may be configured to guide the position to which the battery module **400** is mounted. In one or more embodiments, a plurality of mounting guides **130** may be in the corner portion of the housing **100**. In one or more embodiments, as shown in FIGS. **3** and **4**, two mounting guides **130** may be on each of the left and right corners of the housing **100**. The mounting guide **130** may separate the battery module **400** and the second housing surface **102** such that the corner of the battery module **400** is not in direct contact with the second housing surface **102**. The mounting guide **130** may be configured to align the position of the battery module **400** in response to the battery module **400** being coupled to the housing **100**.

[0096] The housing **100** may further include the contact protrusion **150**. At least one contact protrusion **150** may be in the housing **100**. In one or more embodiments, as shown in FIGS. **3** and **8**, the contact protrusion **150** may be between the first housing surface **101** and the second housing surface **102** and may extend upward (e.g., in the Z-axis direction) along the second housing surface **102** from the first housing surface **101**. In one or more embodiments, the contact protrusion **150** may have a height **H4**. The height **H4** may be the same as or less than the height of the second housing surface **102** and may be the same as or less than the height of a busbar **500**. The contact protrusion **150** may come into contact with the busbar **500** fastened to the battery module **400** as the battery module **400** is inserted into the housing **100**. Therefore, the contact protrusion **150** may closely contact the busbar **500** with the battery module **400**, and thus, an additional component for fixing the busbar **500** does not need to be provided in the battery module **400**.

[0097] The contact protrusion **150** may be between the guide protrusions **450** of the battery module **400**. In one or more embodiments, as shown in FIG. **9**, the contact protrusion **150** may be between a pair of guide protrusions **450** on a side of the battery module **400** and may press the busbar **500** inserted between the pair of guide protrusions **450**. In one or more embodiments, the contact protrusion **150** may be on a horizontal line passing through the center of the connector **600** connecting the busbar **500** to the battery module **400**. In one or more embodiments, the connector **600** may include a conductive material such as a metal and may electrically connect a cell tap to the busbar **500**. The busbar **500** may have a width **W5** and the width **W5** may be the same as (or substantially the same as) an interval or distance between the pair of guide protrusions **450**.

[0098] The contact protrusion **150** may have a tapered shape. In one or more embodiments, the contact protrusion **150** may have a width that narrows in a direction from the bottom to the top (e.g., in the Z-axis direction). In one or more embodiments, as shown in FIG. **8**, a width of a lower end of the contact protrusion **150** (e.g., a width of the housing **100** in the length direction or in the Y-axis direction of FIG. **8**) may be **W4a**, a width of an upper end of the contact protrusion **150** may be **W4b**, and **W4b** may be less than **W4a**. The contact protrusion **150** may include a shape in which a surface facing the battery module **400** is inclined. In one or more embodiments, as shown in FIG. **8**, the surface of the contact protrusion **160** on the opposite side of the second housing surface **102** may face upwards and be tilted to the inside. Therefore, in response to the battery module **400** being inserted into the housing **100**, the bus bar **500** may not interfere with the upper end of the contact protrusion **150** and the battery module **400** may be stably inserted into the housing **100**. The

lower portion of the contact protrusion **150** may press the busbar **500** to closely contact the busbar **500** with the battery module **400**.

[0099] The housing **100** may further include the pocket **160**. At least one pocket **160** may be formed in the housing **100**. In one or more embodiments, as shown in FIGS. **3** and **10**, the pocket **160** may be between the first housing surface **101** and the second housing surface **102** and may be on a side of the second housing surface **102** opposite to the second housing surface **102** on which the contact protrusion **150** is located. The pocket **160** may be configured to hold and support the substrate S. The substrate S may be connected to an LED panel exposed to the outside of the housing **100** to display the current state such as a remaining battery, etc. of the battery pack **10**.

[0100] The pocket **160** may include a barrier rib **161**, a stopper **162**, a first insertion guide **163**, and a second insertion guide **164**.

[0101] The barrier rib **161** may extend from the first housing surface **101** and the second housing surface **102** and may form a space in which the substrate S is accommodated between the first housing surface **101** and the second housing surface **102**. In one or more embodiments, as shown in FIG. **11**, the stopper **162** may extend from the second housing surface **102** toward a space between the barrier rib **161** and the second housing surface **102**. The stopper **162** may be inside the barrier rib **161** and may be configured to guide the position of the substrate S in response to the substrate S being inserted into the barrier rib **161** such that the substrate S does not collide with the second housing surface **102**. In a configuration in which the substrate S is inserted into the barrier rib **161**, the stopper **162** may not be in contact with and may be spaced apart from the substrate S. The stopper **162** may be separated from the inner surface of the barrier rib **161**, the first insertion guide **163**, and the second insertion guide **164**.

[0102] The first insertion guide **163** and the second insertion guide **164** may be configured to fix the substrate S inside the barrier rib **161**. In one or more embodiments, as shown in FIG. **11**, at least one first insertion guide **163** and at least one second insertion guide **164** may be on the second housing surface **102** and may extend toward a space between the barrier rib **161** and the second housing surface **102**. A pair of first insertion guides **163** may face each other on the inner surfaces of the second housing surface **102** and the barrier rib **161**. A separation distance between the first insertion guides **163** on the inner surfaces of the second housing surface **102** and the second insertion guides **164** on the barrier rib **161** may be equal to or greater than a thickness of the substrate S. The substrate S may be inserted and fixed between the first insertion guides **163** formed on the inner surfaces of the second housing surface **102** and the barrier rib **161**. In one or more embodiments, as shown in FIG. **11**, the first insertion guide **163** may include two pairs of first insertion guides **163**.

[0103] At least one second insertion guide **164** may be on the second housing surface **102** and may extend towards a space between the barrier rib **161** and the second housing surface **102**. In one or more embodiments, the second insertion guide **164** may have the same shape as the first insertion guide **163** on the second housing surface **102**. The second insertion guide **164** may be between adjacent first insertion guides **163**, and may be in contact with and thereby support the substrate S. The height of the first insertion guide **163** and the second insertion guide **164** may be less than the height of the second housing surface **102**. The first insertion guide **163** and the second insertion guide **164** may each be inclined or tapered downwards (e.g., in the Z-axis direction). In one or more embodiments, as shown in FIG. **10**, the first insertion guide **163** and the second insertion guide **164** may include an inclined portion that gradually increases in width from the upper end to the lower end and the substrate S may be naturally seated in the barrier rib **161**.

[0104] The external cover **200** may be on the inner cover **300** to cover the inner cover **300** and may be detachably connected to the housing **100**.

[0105] The inner cover **300** may be on the housing **100** and cover the open upper surface of the housing **100**. In one or more embodiments, as shown in FIG. **2**, in a state in which the battery module **400** is accommodated in the housing **100**, the inner cover **300** may be on the battery

module **400** and connected to the housing **100** and the battery module **400** through the fastening member, such as a bolt.

[0106] The inner cover **300** may include a first cover surface **301** and a second cover surface **302**.

[0107] The first cover surface **301** may include an upper surface of the inner cover **300** (in an embodiment in which the external cover **200** is the upper side). In one or more embodiments, as shown in FIGS. **13** and **14**, the first cover surface **301** may correspond to the inner upper surface of the inner cover **300** and may face the battery module **400**. The cover protrusion **310**, the cover rib **320**, and the wing portion **330** may be formed on the first cover surface **301**.

[0108] The second cover surface **302** may include a side surface of the inner cover **300**. In one or more embodiments, as shown in FIGS. **13** and **14**, the second cover surface **302** may extend along an edge of the first cover surface **301** and may correspond to an inner side of the inner cover **300**. The second cover surface **302** may extend in the height direction (e.g., the Z-axis direction of FIG. **14**) from the edge of the first cover surface **301**. The cover protrusion **310**, the cover rib **320**, and the wing portion **330** may be on the second cover surface **302**.

[0109] The inner cover **300** may include the cover protrusion **310** and the cover rib **320**.

[0110] At least one cover protrusion **310** may be on the first cover surface **301** and the second cover surface **302** and the cover protrusion **310** may be connected to the battery module **400**. In one or more embodiments, the cover protrusion **310** may extend in the height direction (e.g., the Z-axis direction of FIG. **14**) from the first cover surface **301** and the second cover surface **302** and may be in communication with an outer upper surface of the inner cover **300**. In one or more embodiments, as shown in FIG. **12**, in a state in which the inner cover **300** is aligned such that the cover protrusion **310** corresponds to the second coupling protrusion **440**, the second coupling protrusion **440** may be inserted into the cover protrusion **310** or the cover protrusion **310** may be inserted into the second coupling protrusion **440**. The cover protrusion **310** may be fastened to the second coupling protrusion **440** by inserting the fastening member, such as the bolt or other fastener, from the outer upper surface of the inner cover **300** into the cover protrusion **310**.

[0111] The cover protrusion **310** may include a first cover boss **311** and a second cover boss **312**.

[0112] At least one first cover boss **311** may be on the first cover surface **301** and connected to the second coupling protrusion **440** of the battery module **400**. In one or more embodiments, as shown in FIG. **13**, a plurality of first cover bosses **311** may be inside the inner cover **300** in the width direction (e.g., the X-axis direction of FIG. **13**) of the inner cover **300** and may be coupled to respective ones of the second coupling protrusions **440**. In one or more embodiments, the plurality of first cover bosses **311** may be arranged in two columns in the length direction (e.g., the Y-axis direction of FIG. **13**) of the inner cover **300** and the first cover bosses **311** included in each column may be connected to different battery modules **400**. In one or more embodiments, the first cover bosses **311** included in each column may be respectively coupled to the second coupling protrusions **440** on the inner longitudinal side of different battery modules **400**. The first cover boss **311** may have a hollow cylinder shape. An end of the first cover boss **311** may be inserted into the second coupling protrusion **440** or may surround the second coupling protrusion **440** and the other end of the first cover boss **311** may be in communication with the outer upper surface of the inner cover **300**.

[0113] The first cover boss **311** may be connected to a plurality of cover ribs **120**. In one or more embodiments, a plurality of cover ribs **320** may extend from the outer circumferential surface of the first cover boss **311**. In one or more embodiments, as shown in FIG. **14**, the plurality of cover ribs **320** (e.g., the first cover rib **321** and the second cover rib **322**) may be arranged at equal intervals (or substantially equal intervals) along the outer circumferential surface of the first cover boss **311**. In one or more embodiments, the plurality of cover ribs **320** (e.g., four cover ribs **320**) may extend in different directions from each other by an angle of approximately 90 degrees from the outer circumferential surface of the first cover boss **311**. The first cover bosses **311** may be respectively connected to each other by the cover ribs **320**. Therefore, the load and vibration applied to the first

cover boss **311** may be distributed through the cover rib **320**.

[0114] At least one second cover boss **312** may be on the second cover surface **302** and be connected to the second coupling protrusion **440** of the battery module **400**. In one or more embodiments, as shown in FIG. **13**, a plurality of second cover bosses **312** may be outside the inner cover **300** in the width direction (e.g., the X-axis direction of FIG. **13**) of the inner cover **300** and be respectively coupled to the second coupling protrusions **440**. In one or more embodiments, the plurality of second cover bosses **312** may be arranged in two columns in the length direction (e.g., the Y-axis direction of FIG. **13**) of the inner cover **300** and the second cover bosses **312** included in each column may be connected to different battery modules **400**. In one or more embodiments, the second cover bosses **312** included in each column may be respectively coupled to the second coupling protrusions **440** on the outer longitudinal side of different battery modules **400**. The second cover boss **312** may have a hollow cylinder shape. An end of the second cover boss **312** may be inserted into the second coupling protrusion **440** or may surround the second coupling protrusion **440** and the other end of the second cover boss **312** may be in communication with the outer upper surface of the inner cover **300**.

[0115] The second cover boss **312** may be connected to the plurality of cover ribs **320**. In one or more embodiments, the plurality of cover ribs **320** may extend from the outer circumferential surface of the second cover boss **312**. In one or more embodiments, as shown in FIG. **14**, the plurality of cover ribs **320** (e.g., the first cover rib **321** and the second cover rib **322**) may be arranged at equal intervals (or substantially equal intervals) along the outer circumferential surface of the second cover boss **312**. In one or more embodiments, the plurality of cover ribs **320** (e.g., four cover ribs **320**) may extend in different directions from each other by an angle of approximately 90 degrees from the outer circumferential surface of the second cover boss **312**. The second cover bosses **312** may be connected to each other by the cover ribs **320**. Therefore, the load and vibration applied to the second cover boss **312** may be distributed through the cover rib **320**.

[0116] The plurality of cover ribs **320** connected to the cover protrusion **310** may extend from different portions of the cover protrusion **310**. In one or more embodiments, as shown in FIG. **13**, the plurality of cover ribs **320** extending from the outer circumferential surface of any one cover protrusion **310** may extend from different portions of the corresponding cover protrusion **310** and may not overlap or be connected to (or at least connected directly to) adjacent cover ribs **320**. Therefore, the impact or vibration transmitted from any one cover rib **320** may be absorbed by the cover protrusion **310** and transmitted to another cover rib **320**, thereby increasing the durability of the cover rib **320**.

[0117] The cover rib **320** may be on the first cover surface **301** and/or the second cover surface **302** and may connect the plurality of cover protrusions **310** to each other. In one or more embodiments, the plurality of cover ribs **320** may extend along the first cover surface **301** and the second cover surface **302** from the circumference of one of the cover protrusions **310** and be connected to an adjacent cover protrusion **310**. The plurality of cover ribs **320** may cross each other to form a grid pattern or a net pattern. The cover rib **320** may be configured to distribute a vibration of the battery module **400** applied to the cover protrusion **310** or an external impact of the battery pack **10** to increase the durability of the inner cover **300** including the cover protrusion **310**. The torque and impact generated in response to the inner cover **300** being fastened to the battery module **400** may be distributed.

[0118] The cover rib **320** may be connected to the second cover surface **302**. In one or more embodiments, as shown in FIGS. **13** and **14**, the cover rib **120** extending from the cover protrusion **310** adjacent to the second cover surface **302** of the housing **100** may be connected to the second cover surface **302**. Therefore, the load and vibration of the battery module **400** may be distributed through the second cover surface **302** through the cover protrusion **310** and the cover rib **320**.

[0119] The cover rib **320** may include a first cover rib **321** and a second cover rib **322**.

[0120] A plurality of first cover ribs **321** may be in the first cover surface **301** and the second cover

surface **302** and may be connected to the cover protrusion **310**. In one or more embodiments, the first cover rib **321** may extend in the width direction (e.g., the Y-axis direction of FIG. **13**) of the inner cover **300** along the first cover surface **301**. The second cover rib **321** may connect the first cover boss **311** to the second cover boss **312** in a horizontal direction. The first cover rib **321** may extend from the outer circumferential surface of the first cover boss **311** and the second cover boss **312** and may cross the second cover rib **322**. The first cover rib **321** may be connected to the second cover surface **302**. In one or more embodiments, as shown in FIGS. **13** and **14**, a portion of the first cover rib **321** extending from the first cover boss **311** and the second cover boss **312** may extend along the first cover surface **301** and contact the the second cover surface **302**. Therefore, the load and vibration transmitted along the first cover rib **321** may be distributed to the second cover surface **302**.

[0121] A plurality of second cover ribs **322** may be on the first cover surface **301** and the second cover surface **302** and may be connected to the cover protrusion **310**. In one or more embodiments, the second cover rib **322** may extend in the length direction (e.g., the Y-axis direction of FIG. **13**) of the inner cover **300** along the first cover surface **301**. The second cover rib **322** may connect the first cover boss **311** and the second cover boss **312** to each other in a vertical direction (e.g., the Y-axis direction). The second cover rib **322** may extend from the outer circumferential surface of the first cover boss **311** and the second cover boss **312** and may cross the first cover rib **321**. The second cover rib **322** may be connected to the second cover surface **302**. In one or more embodiments, as shown in FIGS. **13** and **14**, a portion of the second cover rib **322** extending from the first cover boss **311** and the second cover boss **312** may extend along the first cover surface **301** and contact the second cover surface **302**. Therefore, the load and vibration transmitted along the second cover rib **322** may be distributed to the second cover surface **302**.

[0122] The first cover rib **321** and the second cover rib **322** extending from any one cover protrusion **310** may be connected to different portions of the outer circumferential surface of the corresponding cover protrusion **310**. In one or more embodiments, as shown in FIG. **13**, the first cover rib **321** and the second cover rib **322** may be connected to each of the outer circumferential surfaces of the first cover boss **311** and the second cover boss **312**. The first cover rib **321** and the second cover rib **322** extending from any one first cover boss **311** and any one second cover boss **312** may be connected to different portions of the outer circumferential surface of the corresponding first cover boss **311** and second cover boss **312**. Therefore, the impact or vibration transmitted from any one first cover rib **321** and any one second cover rib **322** may be absorbed by the housing protrusion **110** and transmitted to the first cover rib **321** and the second cover rib **322**, thereby enhancing the durability of the first cover rib **321** and the second cover rib **322**.

[0123] The inner cover **300** may further include the wing portion **330**. In one or more embodiments, as shown in FIG. **14**, the wing portion **330**, in which the cover rib **320** is connected to the cover protrusion **310**, may have a greater height than the cover rib **320** and a lower height than the cover protrusion **310**. In one or more embodiments, as shown in FIG. **14**, the cover rib **320** may be in contact with the outer circumferential surface of the cover protrusion **310** and extend upwards along the height direction (e.g., the Z-axis direction). The wing portion **330** may firmly support a connection portion of the cover rib **320** and the cover protrusion **310**, thereby enhancing the durability of the cover protrusion **310** and the cover rib **320**.

[0124] The wing portion **330** may include a first wing portion **331** and a second wing portion **332**. The first wing portion **331** may be on the outer circumferential surface of the first cover boss **311** and the second wing portion **332** may be on the outer circumferential surface of the second cover boss **312**. The first wing portion **331** may have a greater height than the second wing portion **332**.

[0125] The battery module **400** may include at least one battery cell C and at least one battery module **400** may be included in the battery pack **10**. In one or more embodiments, the battery module **400** may include a plurality of battery cells C and two battery modules **400** may be included in the housing **100**. Each battery module **400** may be connected to the housing protrusion

110 in the housing **100**

[0126] The battery module **400** may include an insertion hole **410**, a first coupling protrusion **420**, a second fastening hole **430**, a second coupling protrusion **440**, and a guide protrusion **450**.

[0127] The insertion hole **410** may be configured to hold and support the battery cell C and a plurality of insertion holes **410** may correspond to the number of battery cells C. The first coupling protrusion **420** may be on the bottom surface of the battery module **400** and inserted into the first housing boss **111**. In one or more embodiments, as shown in FIG. 6, the first coupling protrusion **420** may have a hollow cylinder shape and may have the height h_2 . The first coupling protrusion **420** may be inserted into the first housing boss **111** to be coupled to the housing **100** and the battery module **400**. FIG. 6 shows that the first coupling protrusion **420** is formed under the insertion hole **410**, but embodiments are not limited thereto.

[0128] The second fastening hole **430** may be in the bottom surface of the battery module **400** to correspond to the first fastening hole **140** of the housing **100**. In one or more embodiments, as shown in FIG. 7, the second fastening hole **430** may be inserted into the third housing boss **113**. The second fastening hole **430** may have a height corresponding to the height of the inner surface of the third housing boss **113** (e.g., the height from the inner bottom surface of the third housing boss **113** to the upper end of the third housing boss **113**). The second fastening hole **430** may be in communication with the first fastening hole **140** and may include a screw thread. Therefore, the fastening member, such as the bolt, inserted through the first fastening hole **140** may be coupled to the inside of the second fastening hole **430**.

[0129] At least one second coupling protrusion **440** may be on the battery module **400** and connected to the inner cover **300**. In one or more embodiments, as shown in FIG. 12, a plurality of second coupling protrusions **440** may correspond to the longitudinal side of each battery module **400**. In one or more embodiments, six second coupling protrusions **440** may be in one battery module **400** and three second coupling protrusions **440** may be provided for each longitudinal side of the battery module **400**. The second coupling protrusion **440** may correspond to the cover protrusion **310** of the inner cover **300** and may be connected to the cover protrusion **310** through the fastening member, such as the bolt. In one or more embodiments, the second coupling protrusion **440** may have a hollow cylinder shape and may include a screw thread to allow the fastening member to be fastened inside the second coupling protrusion **440**.

[0130] The guide protrusion **450** may be on the side of the battery module **400** and may fix the busbar **500**. In one or more embodiments, as shown in FIG. 9, two guide protrusions **450** may be spaced apart from each other on the side of the battery module **400**. The busbar **500** may be inserted between the two guide protrusions **450**. In this configuration in which the battery module **400** is seated in the housing **100**, the contact protrusion **150** may press the busbar **500** towards the side surface of the battery module **400**, thereby fixing the busbar **500** to the battery module **400**.

[0131] The battery pack according to the embodiments of the present disclosure may increase the rigidity of the rib supporting the battery module and distribute the load of the battery module, thereby enhancing the durability and reliability of the battery pack.

[0132] It should be understood that embodiments described herein should be considered in a descriptive sense only and not for purposes of limitation. Descriptions of features or aspects within each embodiment should typically be considered as available for other similar features or aspects in other embodiments. While one or more embodiments have been described with reference to the figures, it will be understood by those of ordinary skill in the art that various changes in form and details may be made therein without departing from the spirit and scope of the present disclosure as defined by the following claims.

Claims

- 1.** A battery pack comprising: a housing; and a battery module accommodated in the housing, wherein the housing comprises: a housing protrusion; and a plurality of housing ribs extending in different directions from each other from the housing protrusion, wherein the plurality of housing ribs are spaced apart from each other along a circumferential direction of the housing protrusion.
- 2.** The battery pack as claimed in claim 1, wherein the housing comprises: a first housing surface comprising a bottom surface of the housing; and a second housing surface extending along an edge of the first housing surface in a height direction, wherein each of the plurality of housing ribs is connected to the second housing surface.
- 3.** The battery pack as claimed in claim 1, wherein each of the plurality of housing ribs extending from the housing protrusion is connected to another adjacent housing protrusion of the plurality of housing ribs.
- 4.** The battery pack as claimed in claim 1, wherein the housing protrusion has a hollow cylindrical shape and wherein the housing protrusion is connected to at least a portion of a bottom surface of the battery module.
- 5.** The battery pack as claimed in claim 4, wherein: the battery module comprises a first coupling protrusion on the bottom surface of the battery module, and the first coupling protrusion is in the housing protrusion.
- 6.** The battery pack as claimed in claim 5, wherein: the housing protrusion comprises a first housing boss, the first coupling protrusion is in the first housing boss, and a height of the first coupling protrusion is less than a height of the first housing boss.
- 7.** The battery pack as claimed in claim 4, wherein: the housing comprises a first fastening hole in a bottom surface of the housing, the battery module comprises a second fastening hole in the bottom surface of the battery module, the second fastening hole corresponding to the first fastening hole, and the second fastening hole is in the housing protrusion.
- 8.** The battery pack as claimed in claim 7, wherein the housing protrusion comprises a second housing boss, the second fastening hole is in a third housing boss, and a fastening member extends into the second fastening hole through the first fastening hole.
- 9.** The battery pack as claimed in claim 1, wherein the housing protrusion comprises: a first housing boss; a second housing boss; and a third housing boss, wherein the first housing boss and the third housing boss are connected to the battery module and wherein the second housing boss is not connected to the battery module.
- 10.** The battery pack as claimed in claim 9, wherein the first housing boss and the third housing boss are in the battery module and wherein the second housing boss is spaced apart from a bottom surface of the battery module.
- 11.** The battery pack as claimed in claim 9, wherein the second housing boss is in a center portion of the housing in a width direction of the housing.
- 12.** The battery pack as claimed in claim 9, wherein the third housing boss comprises a plurality of third housing bosses, and the plurality of third housing bosses correspond to a longitudinal side of the battery module.
- 13.** The battery pack as claimed in claim 9, wherein the first housing boss is inside a first area partitioned by a plurality of third housing bosses corresponding to the battery modules and wherein the plurality of housing ribs connect the plurality of third housing bosses.
- 14.** The battery pack as claimed in claim 13, wherein: the first housing boss comprises a plurality of first housing bosses, and the plurality of first housing bosses corresponds to a center portion of the battery module.
- 15.** The battery pack as claimed in claim 13, wherein: the battery module comprises a plurality of battery modules, the second housing boss comprises a plurality of second housing bosses, and the plurality of second housing bosses connect the first housing boss to the plurality of third housing bosses corresponding to different battery modules of the plurality of battery modules.

16. The battery pack as claimed in claim 15, wherein the plurality of second housing bosses are in a second area partitioned by the plurality of third housing bosses corresponding to an outer longitudinal side of any one battery module of the plurality of battery modules, wherein the plurality of third housing bosses correspond to an outer longitudinal side of another battery module of the plurality of battery modules, and wherein the plurality of housing ribs connect the plurality of third housing bosses.

17. The battery pack as claimed in claim 16, wherein: the first area comprises a plurality of first areas, and the second area is between the plurality of first areas and is in contact with each of the plurality of first areas.

18. The battery pack as claimed in claim 2, wherein the plurality of housing ribs comprises: a first housing rib extending in a horizontal direction; a second housing rib extending in a vertical direction; and a third housing rib extending in a diagonal direction between the first housing rib and the second housing rib.

19. The battery pack as claimed in claim 18, wherein the first housing rib, the second housing rib, and the third housing rib are connected to the second housing surface.

20. The battery pack as claimed in claim 1, wherein a height of the plurality of housing ribs is less than a height of the housing protrusion.
